



Modularity and Evolvability of Simian Cranial Sexual Dimorphism

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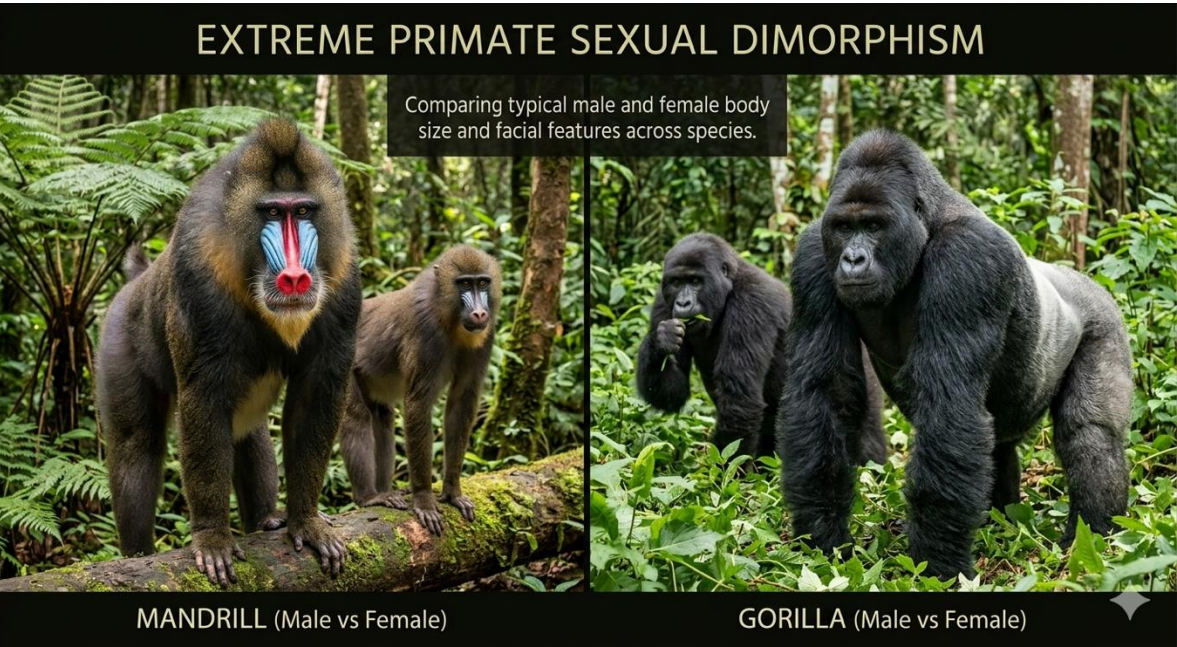
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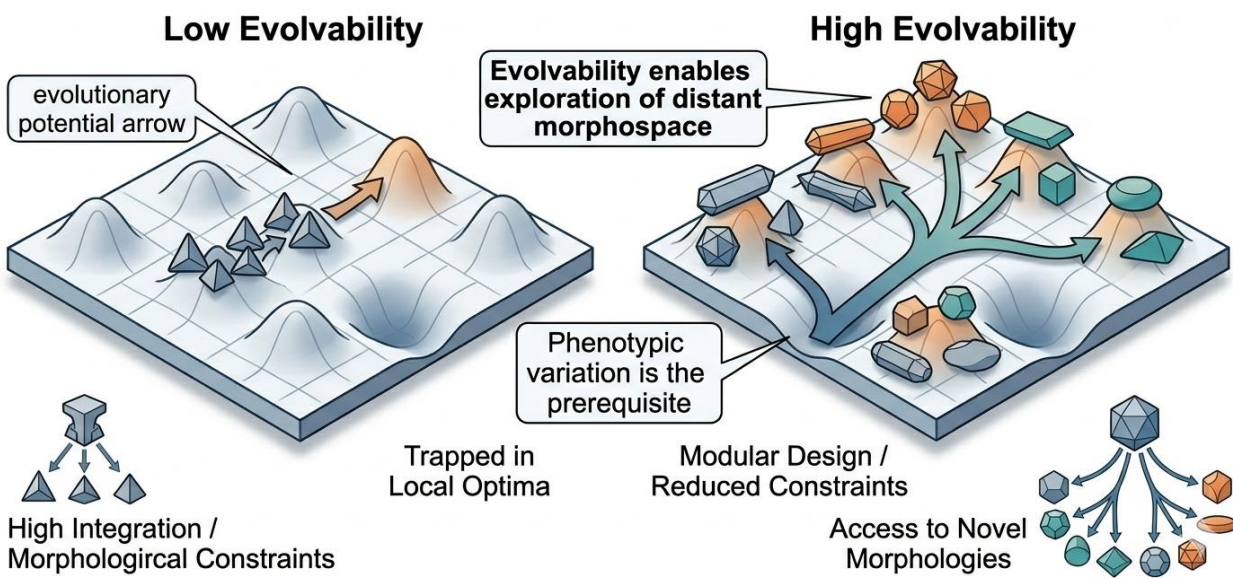
Sexual Dimorphism



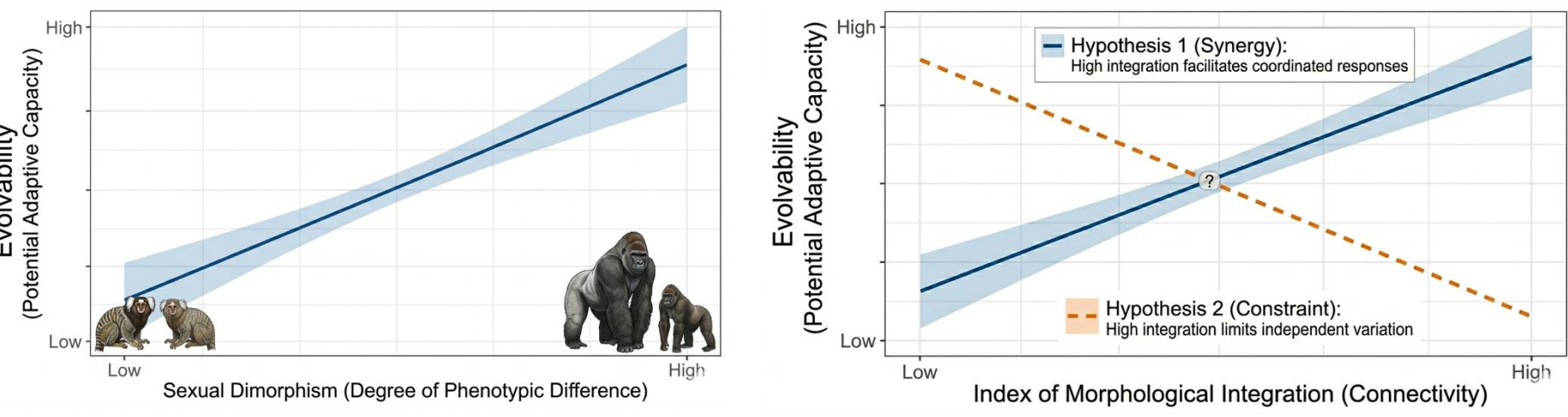
- Sexual dimorphism shows extensive interspecific variation
- The evolution of sexual dimorphism is likely multifactorial, but the role of evolvability remains unclear

Evolvability and Modularity

- Low evolvability may constrain evolutionary change
- High evolvability may promote greater morphological diversification and faster evolutionary change

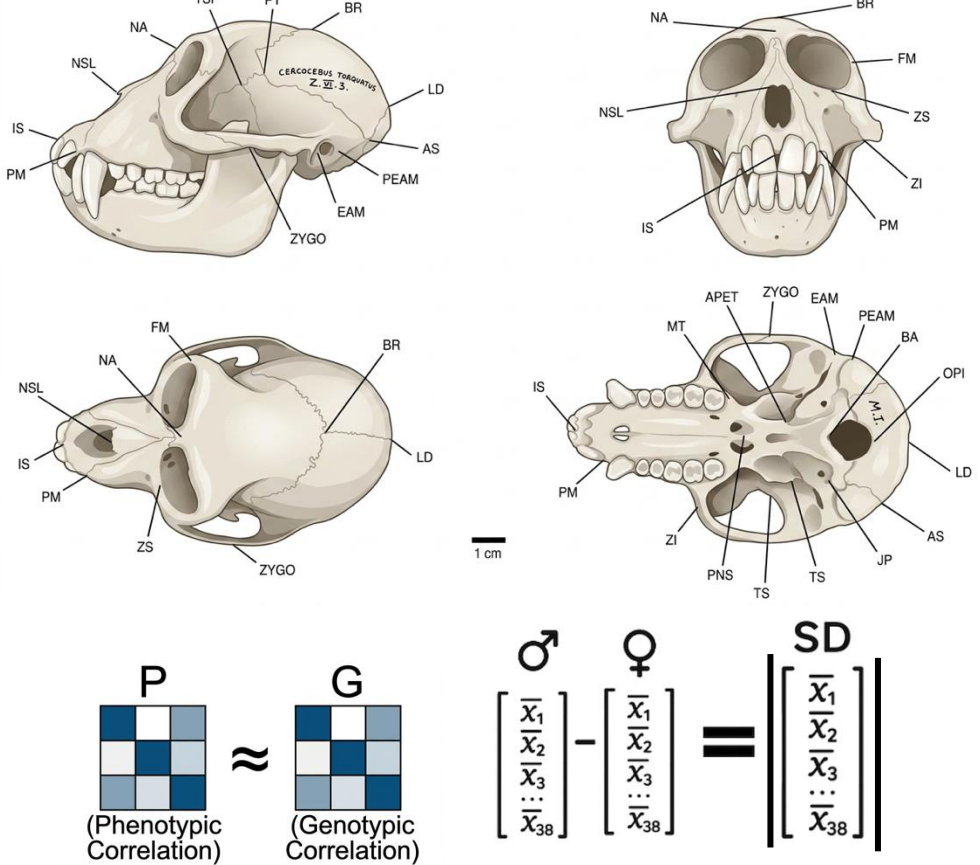


Hypothesis

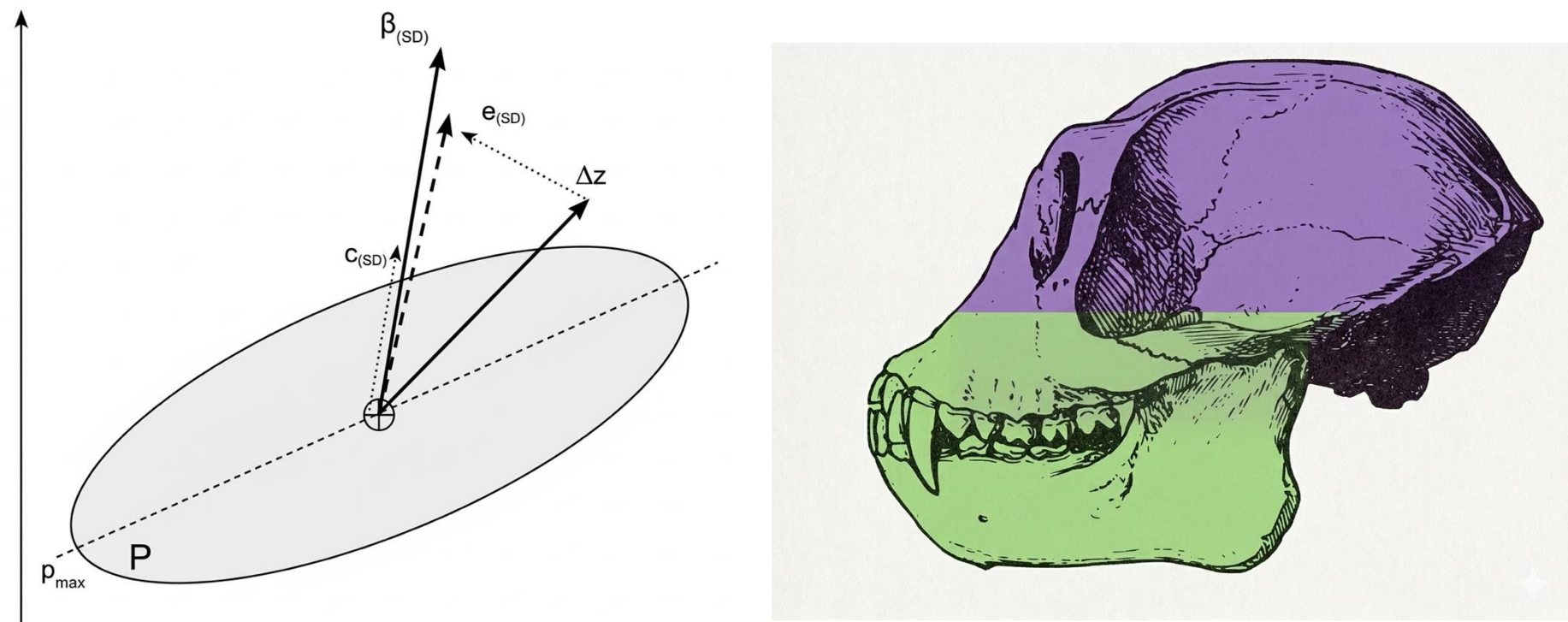


Data

- 50 species from two simian clades: Platyrrhini and Catarrhini
- 38 linear cranial measurements collected from simian skulls
- ~11,000 simian skulls measured.
- P** matrices estimated for each species
- Sexual dimorphism was measured as the **norm** of the vector difference between males and females



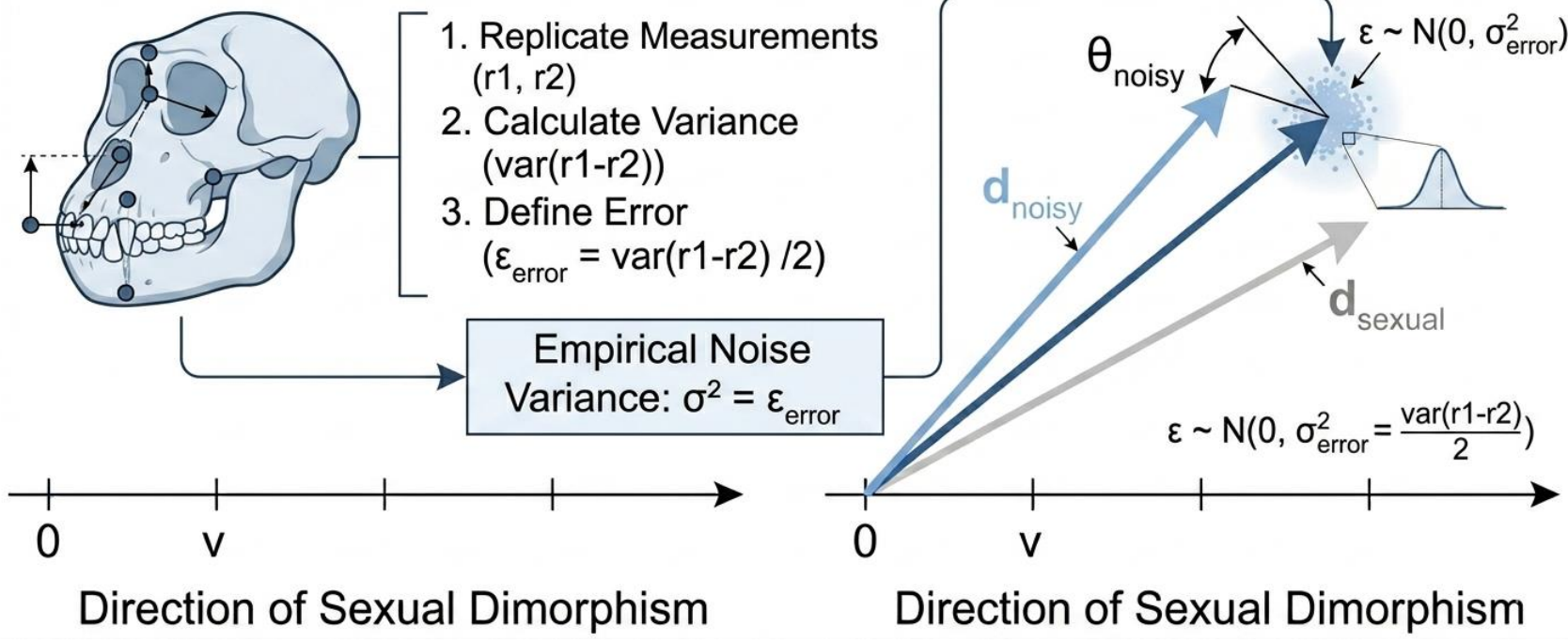
Analysis



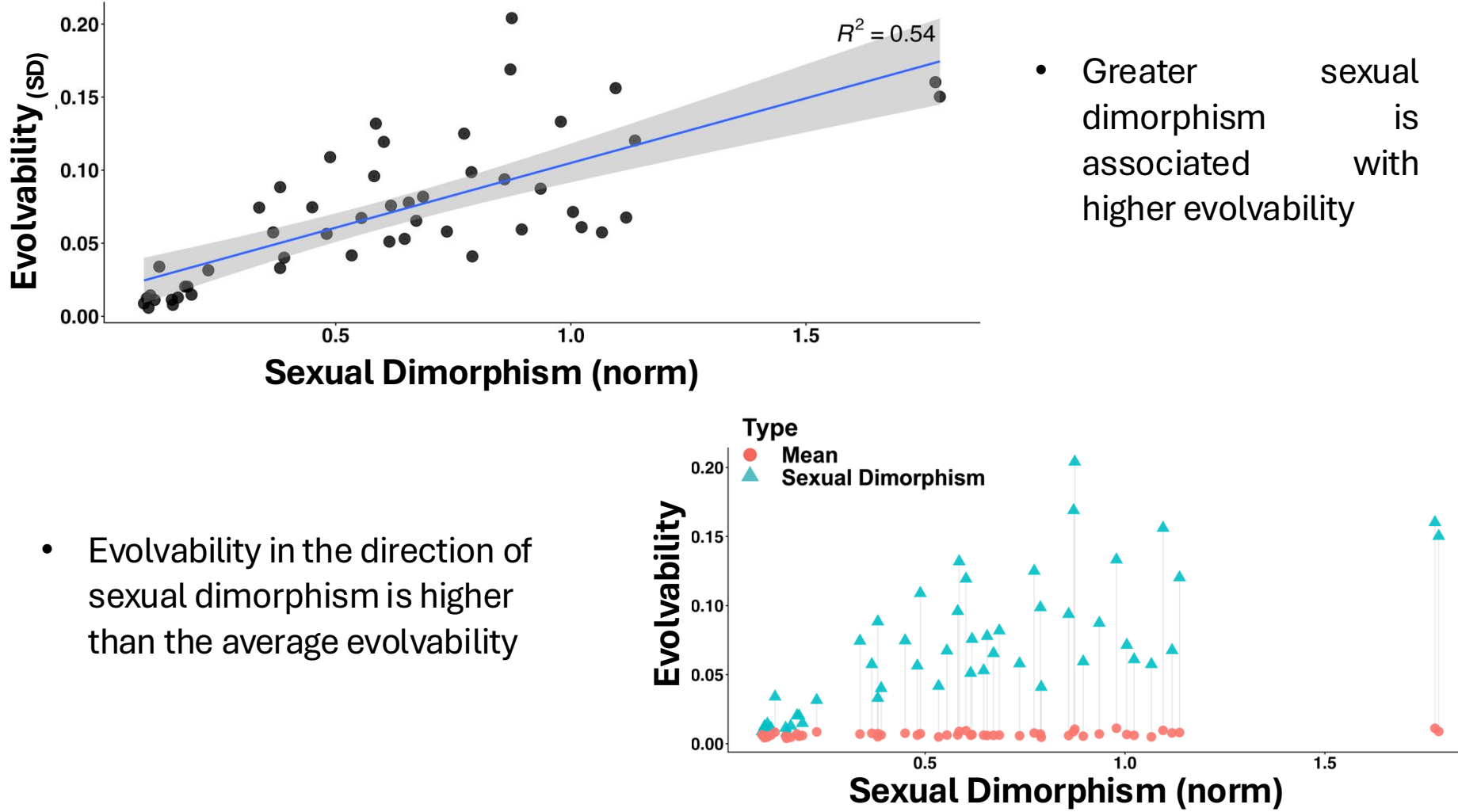
$$\text{evolvability}_{(SD)} = \frac{(SD + \epsilon)' P SD}{|SD + \epsilon|^2}$$
$$\text{integration} = 1 - \frac{1}{SD^T P^{-1} SD}$$

$$\text{conditional evolvability}_{(SD)} = \text{evolvability}_{(SD)} \cdot \text{integration}$$

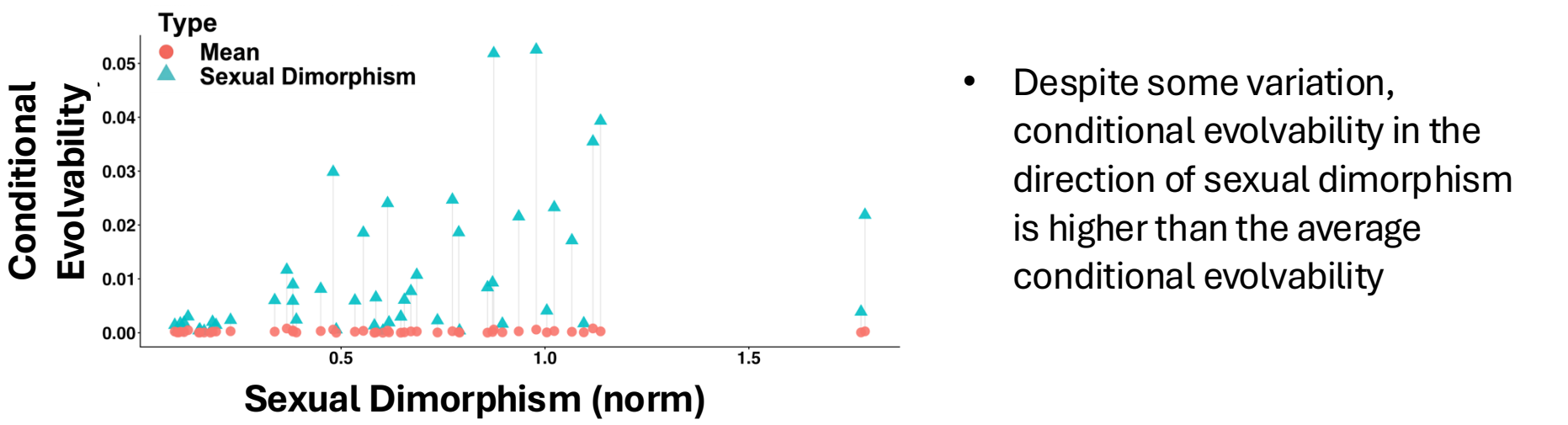
Measurement Error Origin (Primate Crania)



Results

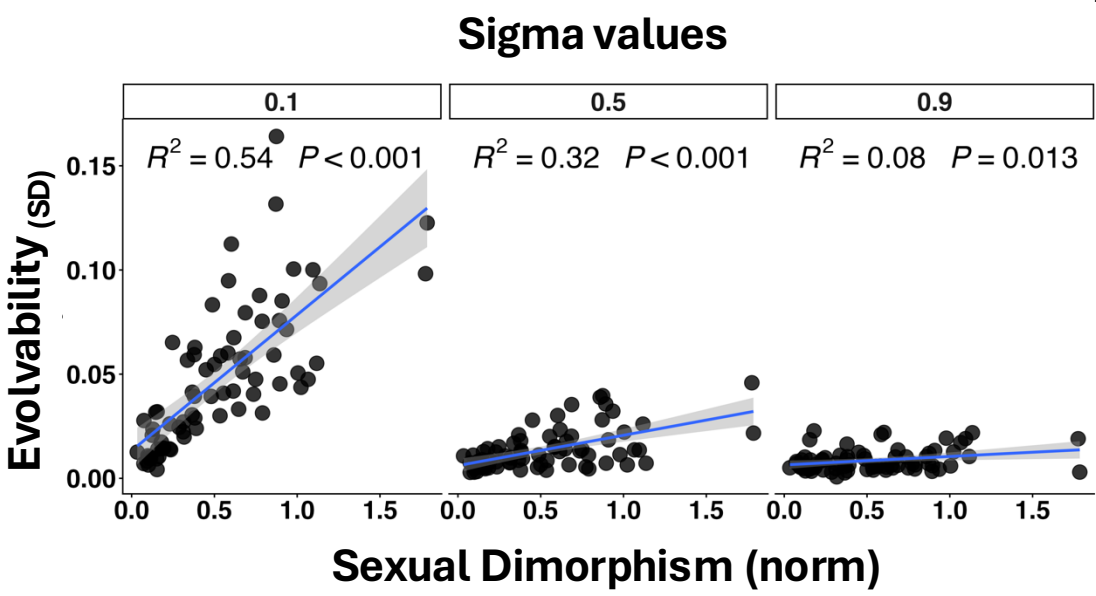
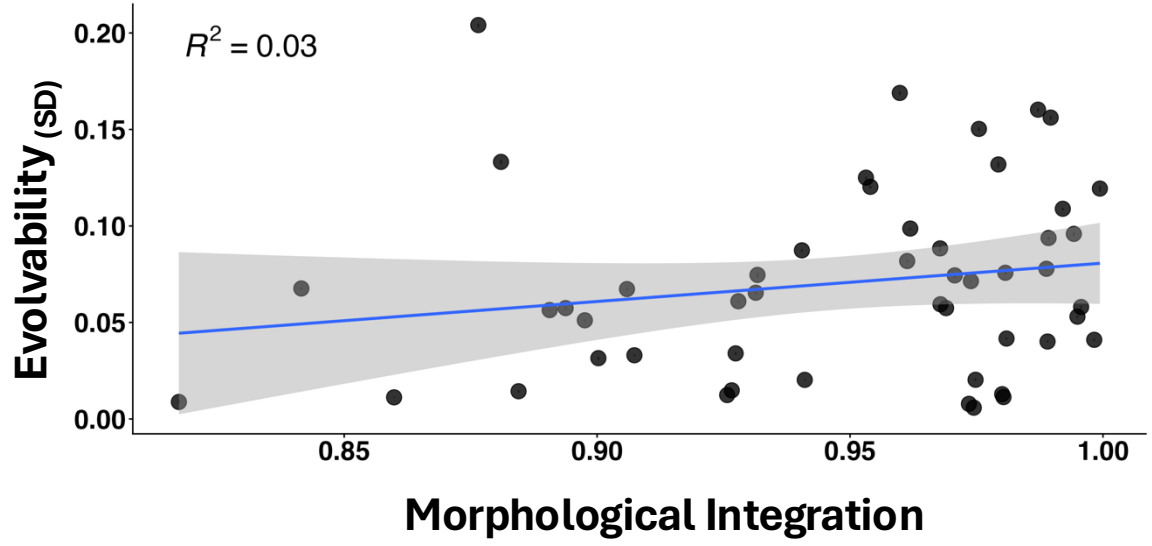


- Evolvability in the direction of sexual dimorphism is higher than the average evolvability

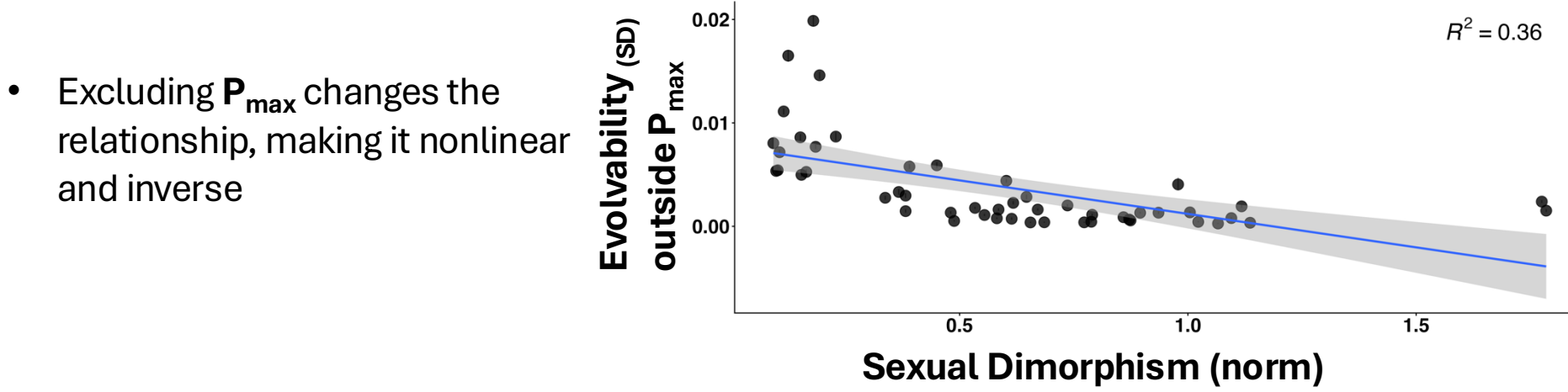


- Despite some variation, conditional evolvability in the direction of sexual dimorphism is higher than the average conditional evolvability

- No relationship was found between evolvability and morphological integration



- High measurement error weakens and may eliminate the relationship between sexual dimorphism and evolvability



Conclusions and Next Steps

- Sexual dimorphism tends to evolve along highly evolvable directions
- Integration alone does not explain variation in evolvability among species
- Measurement error does not substantially alter the observed patterns
- Sexual dimorphism is frequently aligned with **Pmax**
- Identify additional predictors of evolvability in the direction of sexual dimorphism.
- Test whether these patterns are phylogenetically widespread
- Examine the consequences of removing **Pmax** on evolvability estimates.